Here's a **comparison table** between **Docker detached mode** and **interactive mode**:

Feature	Detached Mode (-d)	Interactive Mode (-it)
Purpose	Run container in background	Run container in foreground with user interaction
Command Used	docker run -d <image>	docker run -it <image>
Terminal Access	No terminal access by default	Provides interactive shell access
Use Case	Running background services like web servers, databases	Manual debugging, shell access, or interactive applications
Container Output	Output is not shown in terminal	Output is shown in terminal
Control	Cannot interact directly unless attached later	Can interact using keyboard input
Logs	Use docker logs <container> to see logs	Logs appear live in terminal
Example Command	docker run -d nginx	docker run -it ubuntu /bin/bash
Detach/Exit	Use docker exec or docker attach to connect later	Use Ctrl+P + Q to detach safely; exit to stop container
Automatic Stop	Keeps running until manually stopped	Stops when interactive session ends (e.g., exit command)

Let me know if you'd like a diagram or real-time usage example.

Here's a **real-time usage example** comparing **Docker detached mode (-d)** and **interactive mode (-it)** using an Ubuntu container.

1. Detached Mode Example

Goal: Run an Ubuntu container in the background and create a file inside it.

```
# Run Ubuntu container in detached mode
docker run -d --name ubuntu_detached ubuntu sleep 300
```

- This runs Ubuntu in the background and keeps it alive for 300 seconds using the sleep command.

```
# Check if the container is running
docker ps
# Execute a command inside the running container
docker exec ubuntu_detached touch /testfile.txt
# Check if the file was created
docker exec ubuntu_detached ls /
# Stop the container
docker stop ubuntu_detached
```

2. Interactive Mode Example

Goal: Run Ubuntu in interactive mode and manually create a file.

```
# Run Ubuntu in interactive mode
docker run -it --name ubuntu_interactive ubuntu
```

- This opens a shell prompt inside the container: root@<container-id>:/#

```
# Inside container shell
touch /testfile.txt
ls /
```

- You'll see testfile.txt listed in the root directory.

```
# Exit the container (which stops it)
exit
```

Summary of Behavior

Mode	Shell Access	Background	Output Visible	Auto-Stop on Exit
Detached	✗	✓	✗	✗ (needs stop)
Interactive	✓	✗	✓	✓ (on exit)